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REVIEW ARTICLE

# Soil-health frameworks in agri-food systems. A review

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## Abstract

Soil health is central to agroecological transitions, yet guidance for integrating it into agri-food system design and monitoring remains fragmented. Institutions increasingly use frameworks to define indicators, guide interventions, and report progress against climate, biodiversity, and food-security agendas. However, to our knowledge, there is no integrative soil health framework which coherently links biophysical diagnostics, socio-institutional enablers, and multiscale accountability. This leaves critical gaps in design, sequencing, and measurement of agroecological transitions. Here we review how soil health is operationalized within agroecology and agri-food systems and translate these patterns into an actionable programming guide. We reviewed 64 frameworks and extracted 652 indicators across 12 agroecological principles to build a framework-by-principle evidence matrix. Frameworks were classified by use-orientation (theory, practice, analysis), and indicator thematic profiles were analyzed using hierarchical clustering with adaptive branch detection. The major findings are as follows: (1) framework evolution exhibits four chronological waves with shifts from conceptual foundations to operational measurement and outcome reporting, alongside changes in global and regional agenda setting and a rising demand for comparable indicators; (2) clustering identified five soil health design domains separating biophysical and socio-economic principles and revealing stable micro-constellations beyond earlier pathway framing. These include soil management and input stewardship, soil-health assessment, agroecological and ecosystem-based, integrated landscape and livelihood, and policy- and outcome-based. These findings were translated into a sequenced, multi-domain programming architecture that operationalizes complementarity across diagnostics, stewardship implementation, ecosystem safeguards, landscape–livelihood embedding, and iterative learning, thereby closing the gaps between farm practices, governance mechanisms, and outcome monitoring for soil health.

**Keywords** Soil health · Agroecology · Agri-food systems · Framework · Program design · Implementation · Evaluation · Sustainable development

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## 1 Introduction

The main challenge in agroecological food systems is to balance feeding a growing population while safeguarding environmental integrity (Pinstrup-Andersen and Hazell 1985; Rockström et al. 2009; Godfray et al. 2010; Foley et al. 2011). Soil health is central to this challenge because it underpins agroecosystem productivity and resilience through water and nutrient regulation and carbon storage, with direct implications for climate mitigation, food security, biodiversity conservation, and environmental sustainability (Lal 2004; Kibblewhite et al. 2008; Amundson et al. 2015; IPCC 2019; Madsen et al. 2025). Yet soil degradation now affects over a third of the global agricultural land, driven largely by unsustainable practices, deforestation, and urbanization (FAO-ITPS 2015; IPCC 2019). Beyond biophysical functions, soil health is also foundational to One Health and Planetary Health, including issues such as antimicrobial resistance, zoonotic disease emergence, and terrestrial pollution (Montgomery et al. 2024). These pressures are especially consequential in regions where agriculture underpins livelihoods for most of the population; declining soil health reduces productivity, deepens food insecurity, and increases vulnerability to climate shocks (Sanchez 2002; Amundson et al. 2015; IPCC 2019).

Although the science of soil processes is well established (Figure 1), evidence and guidance are fragmented and insufficiently contextualized to local socio-ecological realities, limiting their translation into actionable interventions and policy design. Science-based frameworks




**Figure 1** Measuring soil infiltration for land degradation surveillance at Linthipe Dedza District in Malawi (Photocredit: Powell Mponela, International Center for Tropical Agriculture (CIAT)).

in agroecological and agri-food system transitions can guide planning, implementation, and evaluation of soil health, but persistent gaps remain in comprehensiveness, inclusivity, and applicability across diverse farming systems (Cowie et al. 2018; Wezel et al. 2020; Utter et al. 2021). While frameworks provide a shared language for stakeholders, their proliferation has generated inconsistencies in scope and design, hindering their collective utility (Hazard et al. 2026). Without coherent integration across biophysical, socio-economic, and governance dimensions, soil health is often monitored as a technical variable rather than being embedded as a strategic design objective within transition processes. As a result, interventions risk remaining project-bound and episodic, limiting cumulative learning, cross-scale coordination, and accountability for long-term soil stewardship outcomes.

Soil health assessment tools offer operational benchmarks but are frequently applied in isolation from broader sustainability planning (Moebius-Clune et al. 2017; USDA-NRCS 2021). Conversely, systems-oriented frameworks increasingly integrate biodiversity, water, livelihoods, climate, and governance, yet often treat soil health indirectly — as a secondary outcome of land-use planning, ecosystem service delivery, or sustainability transitions rather than as a primary design objective (Sanchez 2002; Sayer et al. 2013). For example, FAO's voluntary guidelines provide global practice standards but limited granularity for heterogeneous landscapes (FAO-ITPS 2015, 2017); surveillance approaches can prioritize biophysical metrics while under-addressing socio-economic barriers such as gendered land tenure (Amundson et al. 2015; Utter et al. 2021); and carbon-oriented initiatives can prioritize climate and planetary health benefits while missing near-term agronomic decision needs unless coupled with decision support (Lal 2004; Minasny et al. 2017; Montgomery et al. 2024). These limitations point to a persistent gap between conceptual integration and practical transformation, particularly in dynamic smallholder contexts where scaling from fields to landscapes depends on governance, knowledge systems, and learning under climate and market volatility (Foley et al. 2011; Sayer et al. 2013; Cowie et al. 2018; IPCC 2019).

Agroecology offers a systems lens for repositioning soil health within wider agri-food systems by emphasizing interdependencies among ecological processes, livelihoods, and institutions, and highlighting diversity, resilience, resource efficiency, and the co-creation of knowledge (HLPE 2019; Nicholls et al. 2020; Wezel et al. 2020; Dumont et al. 2021). In this framing, soil health is not only a determinant of production but also a provider of ecosystem services and a foundation for justice, governance, and community connectivity (Sayer et al. 2013; Utter et al. 2021). For instance, for smallholders in fragile environments with limited external inputs, agroecological practices — including nutrient recycling, diversified rotations, and organic amendments — offer practical pathways to regenerate soils and strengthen climate resilience (Sanchez 2002; Tittonell 2014; Altieri et al. 2015; Madsen et al. 2025). The growing literature also presents methodological challenges, as stakeholders struggle to align the perspectives requiring knowledge co-creation and collective learning (Andrieu et al. 2025; Mockshell et al. 2026).

Against this backdrop, the main objective of this review is to examine how soil health is embedded within agroecological and agri-food system framework designs. Specifically, we (i) map how soil health indicators align with the ecological, social, cultural, and economic dimensions articulated by agroecological principles; (ii) consolidate existing frameworks into a comparable evidence matrix to identify design domains; and (iii) translate these domains into a sequenced programming guide that operationalizes their complementarity.

## 2 Framing soil health within agroecological and agri-food systems

Frameworks that address soil health in agroecology and agri-food systems differ widely in purpose, scope, and evidence they provide. This section attempts to establish a common understanding — definitions, principle structure, and synthesis — so that heterogeneous framework families can be compared on a consistent basis and interpreted through the framework-by-principle evidence matrix and the design domains developed in Sections 4 and 5.

### 2.1 Framework definition and institutional strategies

The study adopts the dictionary definition of a framework as "a supporting structure around which something can be built: a system of rules, ideas, or beliefs that is used to plan or decide on something" (Cambridge University Press 2025). The diversity of frameworks observed in agri-food, agroecology, and soil health literature reflects more than disciplinary or methodological differences; it arises from distinct institutional logics and normative orientations that guide how problems are framed, and how solutions are designed. Institutions aim to maximize collective benefit while minimizing costs through frameworks that guide the design of context-relevant interventions (Ostrom 2011). From this perspective, different frameworks emerge and represent competing bundles of theories, practices, and analysis indicators. Institutions are assumed to behave rationally by formulating or selecting the framework that yields the highest utility in terms of policy coherence, operational feasibility, and alignment with its technical, societal, and environmental goals

(Aleskerov et al. 2007; Moullin et al. 2020). However, where indifference exists, multiple frameworks may be treated as functionally equivalent or complementary.

Notwithstanding, the utility-maximization process is not value-neutral. Institutional decisions are also shaped by embedded belief systems and normative commitments, as described by the Value-Belief-Norm (VBN) theory of environmentalism (Oreg and Katz-Gerro 2006). The VBN posits that values influence environmental beliefs, which in turn shape behavioral norms and decisions — such as selecting frameworks that prioritize equity, ecosystem integrity, or inclusive governance. The convergence of utilitarian logic and normative values forms the basis for institution's framework selection, shaping how agroecological transitions are designed, implemented and analyzed. This theoretical framing provides the basis for interpreting patterns observed in the evidence matrix. It reveals that the diversity of frameworks and the uneven representation of agroecological principles are not merely methodological but reflect distinct institutional strategies shaped by both utilitarian reasoning and normative commitments. Framework diversity is not only technical; it reflects institutional strategies shaped by utility-maximizing choices and normative commitments that determine which problems, indicators, and interventions are prioritized. Making these drivers explicit is essential for appraising frameworks in ways that balance operational feasibility with social legitimacy and contextual relevance.

### 2.2 Principles and indicator families used to represent soil health in agroecology

To compare frameworks consistently, the review is organized around agroecological principles and associated indicator families, treating soil health as a key, measurable component that interfaces with other principles (e.g., nutrient cycling, biodiversity, governance, and livelihoods). The principle-based structure primarily draws from agroecological principles and elements (HLPE 2019; Wezel et al. 2020), the tool for agroecology performance evaluation (TAPE) (FAO 2019; Mottet et al. 2020), and the characterization of agroecology transformation (CAET) procedure (Mottet et al. 2020; Lucantoni et al. 2023; Bicksler et al. 2023).

Indicator families were refined using supporting literature that elaborates specific principle areas — including co-creation of knowledge (Utter et al. 2021), human and social values (Bezner Kerr et al. 2022), diversity (Nicholls et al. 2020), animal health (Dumont et al. 2024; Ceppatelli et al. 2025) and soil health measurement guidance (Moebius-Clune et al. 2017; Apfelbaum et al. 2019; FAO-OEWG 2022; Montgomery et al. 2024). This approach enabled a consistent comparison of how frameworks translate principles into measurable priorities. Organizing the review around agroecological principles and indicator families provides a transparent, comparable basis for identifying where soil health is directly operationalized versus indirectly assumed within the broader ecological and socio-institutional priorities.

### 2.3 Review scope and synthesis approach

Literature and grey sources were compiled through structured searching and iterative refinement of keywords to capture diverse framework families spanning agriculture, ecology, and environmental governance. For each framework, indicators were extracted and mapped to the agroecological principle structure (including soil health) to build an evidence matrix used to support both descriptive synthesis and the domain-oriented interpretation.

A Boolean document search was conducted between October 2024 and January 2025 to provide an initial mapping of relevant frameworks. The review was designed to synthesize framework designs (and their indicator emphases) using quantitative structuring (e.g., indicator extraction and clustering) while avoiding statistical inference. To capture frameworks across agriculture, ecology, and environmental sectors, the guiding query combined terms for the focal theme, framework type, and related paradigms: (soil health OR landscape) AND (framework OR analytical framework OR conceptual framework OR theoretical framework OR empirical framework) AND (agroecology OR sustainable intensification OR regenerative agriculture OR conservation agriculture). Searches were conducted in Google Scholar, Scopus, and institutional repositories; query syntax was adapted to platform-specific requirements.

As the compendium expanded, the search was iteratively extended using additional keywords to capture emerging terminology and related framework families. While the full sequence of evolving search strings was not recorded, the Boolean structure above reflects the guiding formulation used throughout the review and provided the basis for mapping framework diversity and situating soil health within wider systems-oriented approaches. Indicators were extracted from documents using the R package pdftools and grouped under the 13 agroecological principles (including soil health). Indicator context was captured by extracting proximal text (extended sentences) around each indicator to support consistent interpretation during mapping and to inform interpretation where principle boundaries are contested.

## 3 Framework evolution

The review uses two complementary lenses to interpret framework diversity. First, frameworks are classified by use-orientation based on established criteria (Rocco and Plakhotnik 2009; UNHCR 2017; Luft et al. 2022): (i) theory-oriented frameworks synthesize concepts to build foundations for theory development; (ii) practice-oriented frameworks conceptualize practices and their associated impacts; and (iii) analysis-oriented ones that provide structured approaches for gathering, analyzing, and interpreting data. This classification clarifies a framework's primary purpose and how it is intended to be used across program cycles.

Beyond use-orientation, the study examines how frameworks differ in their indicator emphasis across the agroecological principle structure (including soil health). To quantify and compare the representation of agroecological principles across diverse frameworks, composite indices were constructed (Nardo et al. 2005) by first applying z-score standardization to indicator counts, followed by unweighted aggregation (mean) at the agroecological-principle level. The constructs were specified reflectively, if observed indicators manifest underlying soil health and agroecological principles (MacKenzie et al. 2011). The study used aggregation and averaging rather than dimensionality reduction methods (e.g. PCA or factor analysis) because these techniques can reduce interpretability and impose statistical assumptions (e.g., orthogonality, linearity) that may not align with multidimensional sustainability constructs (Greco et al. 2019). Aggregation preserves a transparent, additive structure aligned with the conceptual design, supporting clarity for stakeholders and decision-makers (Mazziotta and Pareto 2016).

To interpret framework diversity through pattern analysis, the study applied a row–column hierarchical clustering (simultaneous clustering of frameworks and principles). Clusters were identified using the Dynamic Tree Cut method in R — an adaptive branch-detection algorithm applied to the hierarchical clustering dendrogram that delineates clusters from dendrogram topology and branch stability, rather than imposing a single fixed cut height (Langfelder et al. 2008). The clustering was not used to rank or evaluate frameworks; it summarizes similarities and differences in indicator emphasis to help users identify comparable measurement priorities and assemble complementary framework portfolios that reduce thematic gaps across soil health, agroecology, and agri-food systems, while providing a shared reference for stakeholder communication.

### 3.1 Use-orientation

The review reveals distinctive use-orientations and evolutions of frameworks to support practical selection and adaptation, rather than prescribing a universal approach. From the compendium of 64 agro-food system, soil health and related frameworks, (Figure 2a), almost half focused explicitly on informing either the theory, practice, or analysis; one-fourth focused on two dimensions, and one-tenth on all three. Higher coverage of theories at 68% and analytics at 60%, while only 36% focus on practices, is observed. The explicit focus on theory and analytics signals increased intents to analyze linkages between organizational objectives, design concepts and outcomes, although mere indicator presence may not be related to measurement quality. There is a debate on whether the institutional and program design should focus on the outcomes, interventions, principles or all. Although the common approach has been to prioritize specific interventions (Derpsch 2003; Vanlauwe et al. 2014; Giller et al. 2015; Thierfelder et al. 2018) or adhere to core principles such as sustainability (Pretty 2008), some argue focusing on clearly defined, measurable outputs or outcomes (Musumba et al. 2017). However, the lessons from human health indicate that frameworks do not only aid policy development and tactical decision-making but also jointly guide application of practices and support communication with the larger audience (CCSDH 2015).

Notably, some organizations develop and/or deploy a set of frameworks depending on use, shifting emphasis over time. For instance, the African Union's Comprehensive Africa Agriculture Development Programme (CAADP) developed in 2003 (NEPAD 2003) focused on theory and practice but shifted towards measuring impact with the 2015 CAADP results framework (NEPAD 2015). This temporal pattern reveals four waves in the evolution of framework use-orientation (Figure 2b).

The first wave (1985–2009) established early conceptual foundations around soil-quality and efficiency (Arshad and Martin 2002; Andrews et al. 2004) along with ecosystem service framing (Alcamo et al. 2003; Millennium Ecosystem Assessment 2005) and soil fertility management guidance (Sanginga and Woomer 2009). The second wave (2010–2016) marked a landscape/food-systems transition, shifting from fields to territories with landscape and political economy (Birner and Resnick 2010; Sayer et al. 2013; van der Ploeg 2021). The third wave (2017–2020) reflects growing demand for comparable indicators by formalizing agroecology and landscape guidance (FAO 2017a, 2018, 2019), redefining restoration targets using land-degradation neutrality (Cowie et al. 2018), reframing and synthesis of soil health concepts (Wezel et al. 2020; Lehmann et al. 2020). The fourth wave (2021–2024) emphasized outcome monitoring and corporate alignment, with policy roadmaps and results framework (European Commission 2021; FAO-OEWG 2022), supporting national/continental monitoring guidance (EEA 2023), private-sector buy-in outcome-based tools (Nestle 2024), and expansion of scoring/data

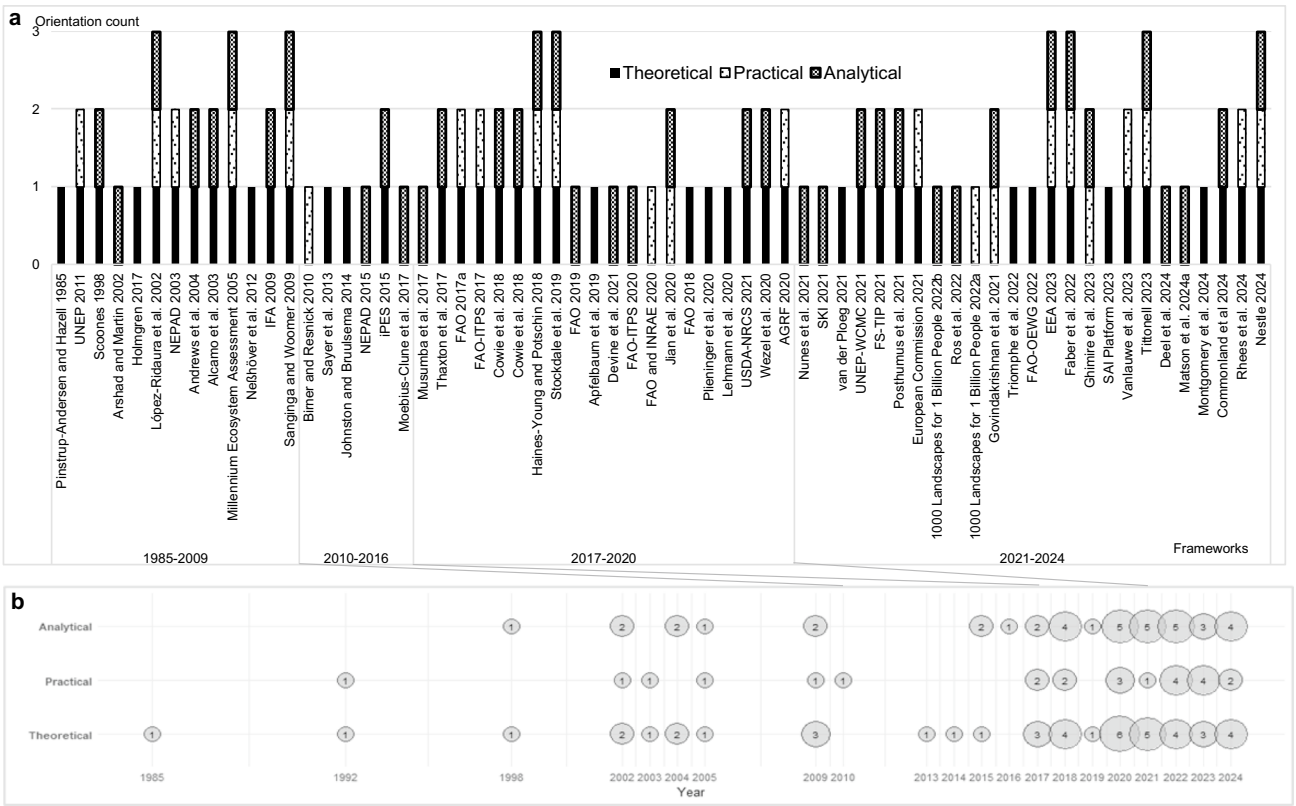



**Figure 2** Use-orientation of reviewed frameworks over time. **a** Frameworks ($n = 64$) ordered chronologically by publication year and classified into three use-orientation categories. **b** Distribution of use-orientation categories by year; circle size reflects the number of frameworks published in that year within each category.

infrastructures (Jian et al. 2020; Deel et al. 2024). Peaks in these years align with climate- and biodiversity-goal setting (e.g., SDGs era), pandemic-era resilience debates, and input-price shocks — further reinforcing attention to monitoring, due diligence, and diversified, landscape-scale solutions. The review traces how framework purposes and indicator emphases shift over time, clarifies why no single framework can serve all decision needs and why selection should be aligned with use cases, scales, and transition stages.

## 3.2 Agroecological principle-orientation

Hierarchical clustering (Figure 3) groups the agroecological principles and frameworks, the dynamic clustering utilizes presence and frequency to determine optimal branching and convergence. Although indicators are not mutually exclusive and vary in relevance as well as clarity and reliability. The column-level clustering further reveals internal branching/convergence structure among agroecological principles within the biophysical and socio-economic clusters.

On the biophysical cluster, the hierarchical clustering yields five sub-branches (i) recycling and biodiversity being closely related, then (ii) animal health, (iii) input reduction, (iv) synergy and (v) soil health. This branching contrasts with Wezel et al. (2020), who aggregated the biophysical principles into two pathways (resource efficiency vs resilience). The dendrogram indicates finer, non-binary structure in which several principles form stable micro-constellations rather than collapsing neatly into the two macro-pathways: recycling and biodiversity share lower branching consistent with the resilience pathway, while animal health, input reduction and synergy branch at incrementally higher node heights — constructed more like cross-cutting enablers, standalone principles.

On the socio-economic cluster: first order constructs aggregate into three pathways: (i) co-creation of knowledge and participation, (ii) fairness, connectivity, and land and natural resource governance, and (iii) economic diversification and social values and diets. Pathways (ii) and (iii) merging at higher nodes may indicate a common governance-market orientation. The study posits that biophysical convergence reflects the historical evolution of evaluation paradigms. Following a prolonged period of productivism, yield-centric evaluation, "second wave" frameworks standardized field-level soil and agronomic metrics — soil

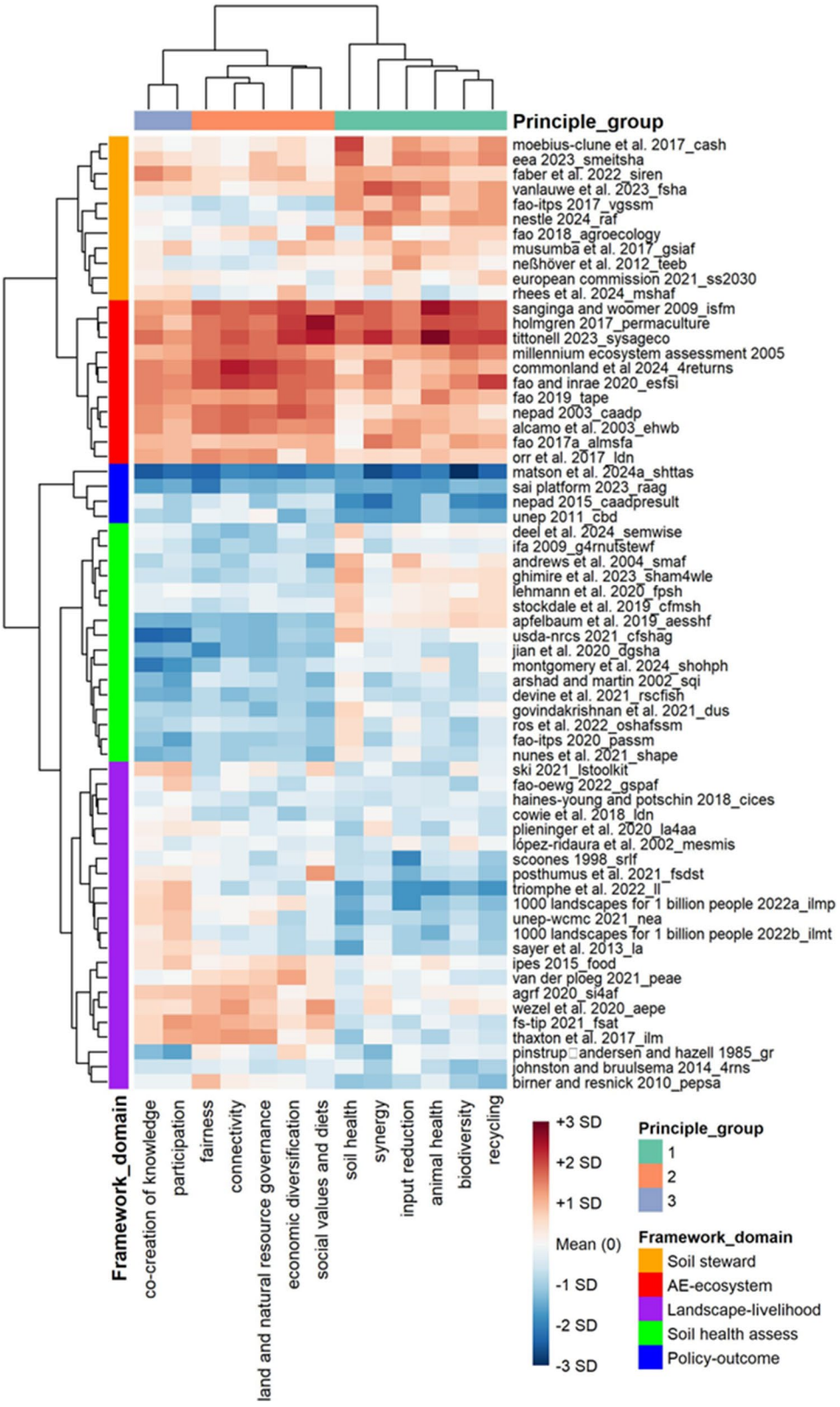



**Figure 3** Z-score heatmap of agroecological principle coverage across frameworks ($n$ = 64). Rows represent frameworks and columns represent agroecological principles. The color gradient shows z-scores of indicator counts at the principle level, where 0 indicates the mean, positive values indicate above-average coverage up to 3 standard deviations (SD), and negative values indicate below-average coverage. Framework clusters and principle groups were delineated from the dendrogram using the Dynamic Tree Cut method. Abbreviated suffixes appended after the author–year citation are used solely for ease of reference and do not represent formal acronyms.

organic matter, erosion control, nutrient use — because they were relatively tractable to measure and had clear plot-scale causal links (Andrews et al. 2004). Socio-economic dimensions were integrated later as attention shifted from plots to farms, landscapes, and food systems, where operationalization required deliberation, institutional

arrangements, and context-specific value choices (HLPE 2019; Wezel et al. 2020). Consequently, most frameworks share a common biophysical framing but diverge along a social–economic one.

This asymmetry has direct implications for design, action and monitoring. Since biophysical "indicators" are broadly similar across the assessed frameworks, strategic distinctiveness — and much of the impact variance — arises from socio-economic differences. Effective designs pair the shared biophysical principles with socio-technical contexts: emphasize co-creation where legitimacy and uptake are binding; strengthen fairness, connectivity, and governance rules where coordination and inclusion are weak; and promote diversification, value addition, and diet-quality incentives where demand and profitability are low (Dumont et al. 2021). Understanding the asymmetry, the stakeholders can be guided to: (i) measure what they design and practice: that is, bundling indicators so each practice set is paired with enabling-environment metrics for scaling; (ii) keep plot-scale agronomic indicators to capture short-run change, and (iii) add governance at farm, landscape, and food-system scales to track sustainability, diffusion, and equity.

## 4 Design domains

The row clustering of Figure 3 delineates five framework design domains — (i) soil management and input stewardship, (ii) soil health assessment, (iii) agroecological and ecosystem-based approaches, (iv) integrated landscape and livelihoods, and (v) policy and outcome-oriented. These five design domains translate complex clustering results into an interpretable typology of how frameworks operationalize soil health within agri-food systems' transitions, supporting clearer selection, combination, and sequencing for planning and evaluation.

### 4.1 Soil management and input stewardship

A cluster of frameworks focuses on farm-level resource efficiency, providing operational guidance and measurement routines — soil health practice guidelines (SOM maintenance, pH or salinity correction, compaction avoidance, infiltration, cover and erosion control) are linked to threshold scoring approaches that make outcomes measurable (FAO-ITPS 2017; Moebius-Clune et al. 2017). Public guidance and standards reinforce monitoring for sustainable soil management (Faber et al. 2022; EEA 2023), with broader strategies framing soil protection to meet regional policy targets (European Commission 2021). Within farms, nutrient stewardship emphasizes right source, right rate, right time and right place (4R) decisions and integrated organic inputs to raise uptake while curbing losses and protecting water quality (Musumba et al. 2017; Vanlauwe et al. 2023), while private-sector frameworks align incentives toward restoring SOM, improving nutrient cycling, and lowering synthetic inputs, with co-benefits for climate resilience and farmer livelihoods (Nestle 2024).

Although soil management and input stewardship frameworks supply indicators to verify efficiency gains, they place less emphasis on the enabling conditions: biodiversity and recycling typically enter as due-diligence items or co-benefits of practice (rotations, cover, and organic amendments) rather than primary design features (Neßhöver et al. 2012). Fairness, connectivity, and animal health appear where standards or reporting instruments are included (FAO-ITPS 2017; Moebius-Clune et al. 2017), but co-creation, participation, and diet or consumption elements are comparatively muted across this stewardship-oriented set. Emphasis remains at micro level; landscape hydrology and cross-farm coordination are rarely integral, limiting their capacity to sustain soil health outcomes beyond plot or farm scale, and require complementing tools to address adoption constraints.

### 4.2 Soil-health assessment

The soil-health assessment frameworks are mixed and bimodal. One pole comprises agronomic toolkits that center on soil health, input efficiency, and management synergies. These include indicator suites, scoring systems, and field protocols that translate the soil triad into operational diagnostics (IFA 2009; Stockdale et al. 2019; Lehmann et al. 2020) — biology (active and total carbon, enzymes, microbial biomass), chemistry (pH, macro- or micro-nutrients), and physics (bulk density, penetration resistance, aggregate stability, available water capacity) — with scoring curves, critical limits, or minimum data sets to flag constraints and guide remediation (Arshad and Martin 2002; Andrews et al. 2004; FAO-ITPS 2020; USDA-NRCS 2021; Govindakrishnan et al. 2021; Deel et al. 2024).

The other pole takes social-ecological and health framing, focusing on fairness or participation, connectivity, and diversification while treating biophysical metrics as contextual. Regional or context-specific frameworks and open databases enable adaptation monitoring and evaluation (Jian et al. 2020; Devine et al. 2021; Ros et al. 2022), with variants focused on water-limited environments and soil organic carbon (Nunes et al. 2021; Ghimire et al. 2023). Recent studies position soil health within one-health or planetary-health agendas, broadening the purpose of diagnostics beyond agronomy (Lehmann et al. 2020; Montgomery et al. 2024). Much as assessment frameworks offer a diagnostic spine (protocols, thresholds, scoring, datasets) that can anchor monitoring and adaptive management.

### 4.3 Agroecological and ecosystem-based

Agroecological and ecosystem-based frameworks pivot from plot-level practices to landscape and system governance, emphasizing co-creation or participation, connectivity and tenure or governance, and multifunctional outcomes (FAO 2018). They couple agroecological practice with territorial coordination, ecosystem-service accounting, and participatory performance tools, moving the locus of change from fields to food systems. One subset focuses on landscape articulation of spatial coordination through stakeholder platforms (FAO 2017b, a), restoration and long-term value creation (Commonland 2024), and system-level feedback (Tittonell 2023). The food-systems sub-block locates farm practices within enabling policy and market settings (FAO and INRAE 2020) and within regional development agendas (NEPAD 2003). Another subset consolidates environmental targets to anchor soil and land stewardship in accounting and planning (Cowie et al. 2018), while ecosystem-service framings provide the benefits ledger and trade-off criterion for design and evaluation (Alcamo et al. 2003; Millennium Ecosystem Assessment 2005). Participatory performance tools operationalize co-creation and learning — from plot to biomes — through context-specific diagnostics and indicators (FAO 2019).

Relative to management and stewardship frameworks, soil functions are secured indirectly — through diversity, ground cover, water regulation, and circular nutrient flows. Agroecological and ecosystem-service approaches promote permanent cover, intercropping, N-fixing species, integrated crop–livestock systems, and reuse of organic resources (Alcamo et al. 2003; Holmgren 2017; FAO 2019). Landscape instruments strengthen erosion control and hydrological regulation via mosaics of land uses, riparian buffers, terraces, and restoration zones, and they institutionalize cross-farm coordination (NEPAD 2003; FAO 2017a; Cowie et al. 2018; Commonland 2024). Within this domain, farm-level nutrient guidance is retained but is explicitly embedded in system design — as one lever among diversification, agroecological redesign, and social organization (Sanginga and Woomer 2009; Tittonell 2023). Biodiversity, recycling or circularity, and land and natural-resource governance dominate, while farm-scale input-reduction and within-farm "synergy" register weakly.

Fairness, connectivity, tenure, and coordination are expressed through collective rules and planning (NEPAD 2003; Cowie et al. 2018; FAO and INRAE 2020), co-creation or participation appears where participatory performance tools structure co-learning across scales (FAO 2019), while diet or consumption elements are secondary and often contingent on food system policies rather than field protocols (FAO and INRAE 2020).

Agroecological and ecosystem-service frameworks embed soil health within territorial governance, coordination, and multifunctionality, shifting the locus of change from field practices to systems organization. Their strength is institutionalizing collective action and landscape integrity, while detailed agronomic routines are typically delegated to complementary toolkits.

### 4.4 Integrated landscape and livelihood

Emerging, this domain groups broad, integrative frameworks that address multiple dimensions (environmental, social, economic) and multiple scales (from farm to landscape to national) in a systems perspective. These frameworks fuse food-system indicator sets with landscape or governance instruments. Agriculture and natural resource management are considered as part of a complex socio-ecological system — factoring in livelihoods, governance, equity, and long-term system change. The unifying characteristic is a holistic view that bridges traditional sectoral silos, aiming for simultaneous progress in productivity, poverty reduction, and environmental sustainability.

Despite the scaling potential, it is worth noting that the geographic coverage and intended applications may vary widely. Frameworks originating in the Global North can encode institutional assumptions distinct from those co-designed in smallholder contexts. Co-creation, participation, fairness, and connectivity are built in via livelihoods, food-systems and political economy lenses (Scoones 1998; López-Ridaura et al. 2002; iPES 2015; Food System Transformative Integrated Policy (FS-TIP) initiative 2021; Posthumus et al. 2021; van der Ploeg 2021; Triomphe et al. 2022). Land- and natural-resource governance is formalized through integrated landscape management and restoration planning (Sayer et al. 2013; Cowie et al. 2018; SKI 2021; 1000 Landscapes for 1 Billion People 2022a, b). Biodiversity and ecosystem-service accounting anchor benefit accounting and monitoring benchmarks (Haines-Young and Potschin-Young 2018; UNEP-WCMC 2021). Diets and economic diversification appear through food-systems analysis and agroecology reviews rather than design features (iPES 2015; Wezel et al. 2020; Food System Transformative Integrated Policy (FS-TIP) initiative 2021). Animal health is sparsely covered (Johnston and Bruulsema 2014; AGRF 2020; FAO-OEWG 2022; Rhees et al. 2024).

Soil outcomes are also secured indirectly — via diversified land use (e.g., agroforestry), ground cover at scale, reduced soil erosion, and improved infiltration — while technical specifics (pH or salinity correction, traffic management, nutrient timing) are typically delegated to complementary agronomic toolkits (Sayer et al. 2013; Plieninger et al. 2020). Integrated frameworks bridge scales and dimensions to support transformation agendas, but their applicability

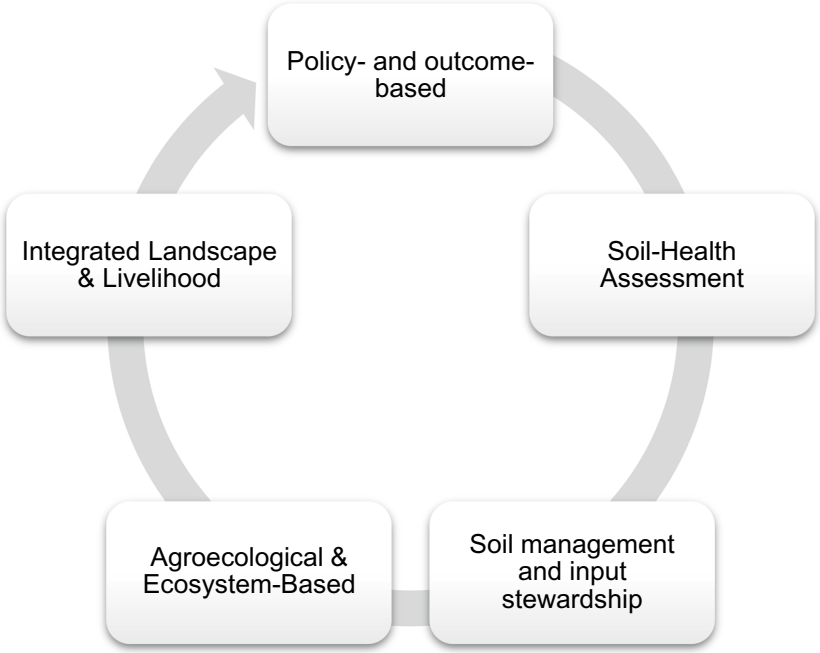



**Figure 4** Sequenced integration of the five framework design domains within an adaptive programming cycle to support iterative learning.

depends on contextual fit and often requires coupling with soil-specific diagnostic and management tools to translate broad objectives into actionable soil stewardship.

### 4.5 Policy and outcome-orientation

The high-level policy frameworks and outcome-based standards set targets, indicators, or broad principles to guide sustainable agriculture and resource management. They are often driven by governments, international bodies, or industry coalitions and serve as "reference frameworks" for aligning strategies and monitoring progress toward sustainability goals. The common thread is establishing measurable goals or commitments — whether for soil health, biodiversity, or agricultural performance — and creating accountability through indicators and reporting. Policy and convention instruments provide the anchor for tracking global obligations, continental agendas and national commitments while sectoral standards translate these into soil health target or threshold and regenerative outcome metrics (NEPAD 2011; UNEP 2011; SAI Platform 2023; Matson et al. 2024).

Soil health is framed via high-level targets, thresholds, or minimum safeguards (e.g., indicator "floors" for organic carbon, erosion risk, or basic biodiversity). The emphasis is on measurable goals, indicators, and reporting — mainstreaming biodiversity and soil health through key performance indicators — rather than on detailed, farm-level practice and measurement guidelines (Pinstrup-Andersen and Hazell 1985). In this configuration, soil health, input reduction, and biodiversity function mainly as baseline safeguards (e.g., indicator floors); recycling, synergy, co-creation or participation, and consumption-side elements are largely absent. The domain strengthens accountability through targets and comparability yet risks superficial compliance if not linked to diagnostics and practice bundles. The greatest value is the enabling monitoring–reporting–verification through which soil stewardship can be incentivized and aligned across programs.

## 5 Integration and programming

Content analysis and clustering indicate that frameworks tend to be designed following complementarity, not substitution. Metrics increasingly converge around shared biophysical and agroecological principles, which creates a stronger basis for more coherent and integrated assessment systems (Dumont et al. 2016). Historically, framework development appears to have evolved in response to emerging needs: from an initial emphasis on soil health measurement and management, to landscape-based scaling, and more recently to outcome-, governance- and stewardship-oriented approaches. Rather than representing competing paradigms, the framework clusters can be understood as complementary design domains, each addressing distinct constraints encountered during farming-system transition. Building on these findings, we propose a target-based programming guide that operationalizes complementarity across clusters and domains, positioning soil health as the central organizing principle within adaptive, multi-scale transition design (Figure 4).

The guide is intended to support stepwise programming rather than a one-framework-per-domain selection exercise. For a given intervention, the process begins by defining the purpose of the assessment, the decision context, and the spatial scale. Policy- and outcome-oriented frameworks can be used to set measurable goals, performance criteria, and accountability mechanisms. At this stage, users should identify a limited number of critical points that either weaken or strengthen soil health, drawing on literature review, expert consultation, and participatory diagnosis. This first filtering step is important to avoid an overabundance of metrics and to focus only on the most decision-relevant domains and indicators.

Once priorities are identified, stakeholders can select one or several frameworks within each relevant domain and extract candidate indicators from them. The aim is not to adopt entire frameworks wholesale, but to use them as structured sources of indicators, concepts, and measurement logic. Indicators can then be screened for relevance, feasibility, interpretability, and redundancy, producing a parsimonious set adapted to the intervention context. In this way, the process moves across domains while retaining only those elements that contribute directly to the assessment objective.

Soil health assessment frameworks are particularly useful at this stage for establishing baseline diagnostics and standardized measurement protocols. The methods and tools used to assess soil health and related agroecological indicators should be adapted to local conditions and translated into forms that are meaningful to a wider audience, including farmers, extension agents, and local decision-makers. Soil stewardship and management frameworks can then be used to identify and test practice options and operational routines. Rather than prescribing a fixed intervention package, the compendium and decision matrix should be used as a basket of options from which context-appropriate combinations can be selected. Field and farm segmentation is important here, because different practices may be suitable for different farming systems, resource endowments, and landscape units.

For example, in a maize-based smallholder system, programming might combine: (i) condition indicators such as soil organic matter, infiltration, and nutrient balance from a soil health assess framework; (ii) management indicators such as residue retention, crop rotation, or organic amendments from a soil steward framework; (iii) biodiversity and resilience indicators from an AE-ecosystem framework; and (iv) equity, market access, or governance indicators from landscape-livelihood and policy-outcome frameworks. These indicators would then be consolidated into one context-specific monitoring set aligned with the intervention's goals and practical data constraints.

Scaling should then be organized within agroecological and ecosystem boundaries so that adoption does not undermine biophysical integrity or intensify trade-offs as interventions expand. Local scaling partners play a central role in co-designing this strategy. Because landscape boundaries are nested, overlapping, and socially differentiated, integrated programming requires frameworks that address both ecological and social dimensions. The compendium therefore supports not only biophysical targeting, but also the integration of governance, market linkages, and livelihood objectives.

Although Figure 4 presents a sequenced learning cycle, the process is not strictly linear. Spatial scales are nested or overlapping, and feedbacks occur through shorter iterative loops of learning, course correction, and adaptation. Assessing the relevance, feasibility, and accuracy of selected metrics and indicators should therefore be part of the adaptive learning process itself. By reframing frameworks as complementary design domains linked through an adaptive programming cycle, research-for-development can operationalize soil health as a systems-level organizing principle rather than as a stand-alone assessment category.

## 6 Conclusion

Soil health is widely recognized as an important biophysical and socio-economic dimension of agroecological and agri-food system transitions, yet the proliferation of frameworks has contributed to fragmented guidance and uneven integration across assessment and design efforts. This review offers a comparative structure for examining heterogeneous framework families and for considering how institutions position soil-related indicators within broader system priorities. Across the compendium, variation among frameworks appears to reflect not only methodological differences, but also institutional strategies shaped by utility considerations and normative orientations that influence problem framing, indicator selection, and the balance between technical performance and social legitimacy.

The analysis suggests that many frameworks converge around a set of shared biophysical principles, while showing greater variation in socio-economic dimensions. This pattern may indicate that differences in governance, inclusion, learning processes, and market conditions are particularly important in shaping how soil health is interpreted and operationalized in practice. The five design domains proposed here translate these patterns into a working typology. Stewardship and assessment domains tend to support practice guidance and diagnostic routines; agroecological or ecosystem, and integrated landscape or livelihood, domains help situate scaling within territorial and governance contexts; and policy or outcome domains can help frame targets, accountability, and comparability across monitoring and reporting systems.

Because these domains appear to address different but related constraints, soil stewardship may be better supported through sequenced combinations of frameworks

than through reliance on a single framework alone. On this basis, instead of proposing another standalone framework, the review outlines a programming pathway that begins with policy or outcome targets, links them to soil health diagnostics, uses stewardship-oriented frameworks to inform practice options, considers scaling within agroecological or ecosystem boundaries, extends analysis to landscape and livelihood contexts, and returns to diagnostics through iterative learning. This sequencing is intended as a practical way of using complementarity across frameworks and may help position soil health more centrally within efforts to support resilient, equitable, and sustainable agri-food system transitions.

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**Data Availability** The indicator terms and datasets extracted and/or analyzed during the study are available in the Zenodo repository, https://zenodo.org/records/17453617.

**Code availability** The R code used for PDF parsing, term proximity extraction, framework-by-principle evidence matrix construction, and comparative framework analysis is available on the corresponding author's GitHub repository, https://github.com/powellmponela/soilhealth_ae_framework_review.

## Declarations

**Conflict of interest** The authors declare no competing interests.

**Ethics approval and consent** Not applicable.

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